

The endomorphic condition $5''_s$ as an algebraic characterization of constant-Jacobian endomorphisms

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Abstract

We present a new algebraic approach to polynomial endomorphisms with constant Jacobian, based on the endomorphic condition $5''_s$ arising from the theory of square-free factorizations. This condition describes the behaviour of square-free divisors in factorizations $a = bc$ and turns out to be equivalent to the preservation of square-freeness by constant-Jacobian polynomial endomorphisms. We show that for such endomorphisms the condition $5''_s$ is equivalent to being an automorphism.

Using Drużkowski's reduction, we reduce the analysis of constant-Jacobian endomorphisms to the study of cubic forms of the shape $F(x) = x + (Bx)^3$ with nilpotent matrix B . We prove that the condition $5''_s$ is local, stable under composition, and invariant under automorphisms, which allows it to be carried through the entire Drużkowski reduction. We obtain the equivalence: a constant-Jacobian endomorphism is an automorphism $\iff$ its Drużkowski form satisfies $5''_s$.

The analysis of Drużkowski forms in low dimensions and for Jordan blocks leads to a new, purely algebraic characterization of the class of constant-Jacobian endomorphisms, reducing their automorphicity to the control of square-free divisors in the images of cubic forms.

1 Introduction

The Jacobian Conjecture, formulated by O. Keller in 1939 [10], is one of the most important problems in the algebraic theory of polynomial mappings. In its classical form it states:

If a polynomial endomorphism

$$\varphi : k[x_1, \dots, x_n] \rightarrow k[x_1, \dots, x_n],$$

over a field of characteristic 0 has constant Jacobian $\det J(\varphi) \in k^*$, then φ is an automorphism.

Despite intensive research, a complete solution to the Jacobian Conjecture remains unknown. In particular, numerous reductions are known which reduce the problem to increasingly specific classes of endomorphisms. The most important of these is Drużkowski's reduction, which shows that in the study of constant-Jacobian endomorphisms one may restrict attention to mappings of the form

$$F(x) = x + (Bx)^3,$$

where B is a nilpotent matrix. These mappings, called *Drużkowski cubic forms*, play a central role in the analysis of the structure of constant-Jacobian endomorphisms.

The aim of this work is to present a new, purely algebraic approach to the study of constant-Jacobian endomorphisms, based on the analysis of square-free divisors and on the endomorphic condition $5_s''$. This condition, introduced in [11, 12], describes a precise control over square-free divisors in factorizations of elements and turns out to be particularly powerful in the context of polynomial mappings.

The central result of the paper is the following:

The endomorphic condition $5_s''$ is equivalent to being an automorphism for constant-Jacobian endomorphisms.

In particular:

- if the image of an endomorphism satisfies $5_s''$, then the endomorphism is an automorphism,
- the condition $5_s''$ is local,
- the condition $5_s''$ is stable under composition,
- the condition $5_s''$ is invariant under automorphisms,
- Drużkowski cubic forms with nilpotent matrix always satisfy $5_s''$.

Combining these facts yields a new algebraic characterization of constant-Jacobian endomorphisms:

a constant-Jacobian endomorphism is an automorphism $\iff$ its Drużkowski form satisfies $5_s''$.

The paper consists of the following parts:

1. Introduction.
2. Preliminaries: definitions of square-free elements and the conditions 5_s , $5'_s$, $5''_s$.
3. Properties of the endomorphic condition $5''_s$: locality, stability under composition, invariance.

4. Druzkowski cubic forms: analysis of square-freeness and the condition $5_s''$.
5. Dimension 2: complete classification of cubic forms and proof of the condition $5_s''$.
6. Dimension 3: analysis of diagonal, triangular, and nilpotent matrices.
7. Nilpotent matrices in arbitrary dimension: analysis of Jordan blocks and their sums.
8. A proof scheme in the language of the condition $5_s''$.
9. Consequences and closure of the scheme.

As a result, we obtain a compact algebraic characterization of the class of constant-Jacobian endomorphisms for which the condition $5_s''$ holds.

2 Preliminaries

In this section we present the basic definitions concerning monoids, square-free elements, and the divisor conditions $5_s, 5'_s, 5''_s$. We then pass to their endomorphic counterparts in the polynomial ring $A = k[x_1, \dots, x_n]$. These notions form the foundation for the subsequent parts of the paper.

2.1 Square-free elements in monoids

Let H be a commutative monoid. By $\text{Sqf}(H)$ we denote the set of *square-free* elements, i.e. those that have no non-unit square divisors:

$$a \in \text{Sqf}(H) \iff \forall d \in H (d^2 \mid a \Rightarrow d \in H^*).$$

This definition and its properties originate from [8, 11, 12]. Square-free elements correspond to reduced principal divisors in geometric language.

2.2 Divisor conditions $5_s, 5'_s, 5''_s$

The conditions $5_s, 5'_s, 5''_s$ describe various forms of control over square-free divisors in factorizations $a = bc$.

Condition 5_s . A monoid H satisfies condition 5_s if for every $a \in H$ there exist $b \in H$ and $c \in \text{Sqf}(H)$ such that

$$a = bc \quad \text{and} \quad a \mid c^m \text{ for some } m \in \mathbb{N}.$$

Condition $5'_s$. A monoid H satisfies condition $5'_s$ if for every $a \in H$ there exist $b \in H$ and $c \in \text{Sqf}(H)$ such that $a = bc$, and

$$d \mid a, d \in \text{Sqf}(H) \Rightarrow d \mid c.$$

Condition $5_s''$. A monoid H satisfies condition $5_s''$ if for every $a \in H$ there exist $b \in H$ and $c \in \text{Sqf}(H)$ such that $a = bc$, and

$$d \mid b, d \in \text{Sqf}(H) \Rightarrow d \mid c.$$

Condition $5_s''$ is the strongest in this hierarchy. It possesses stability, locality, and invariance properties, which we discuss in later sections. We emphasize that the monoid conditions 5_s , $5'_s$, $5''_s$ concern factorization structure in a monoid and are not identical to their endomorphic counterparts in polynomial rings.

2.3 Endomorphic condition $5''_s$

Let $A = k[x_1, \dots, x_n]$ be a polynomial ring over a field of characteristic 0, and let $\varphi : A \rightarrow A$ be a k -algebra endomorphism. We denote its image by $R = \varphi(A)$.

Definition 2.1 (Endomorphic condition $5''_s$). *The image $R = \varphi(A)$ satisfies the endomorphic condition $5''_s$ if for every $f \in A$ and every factorization*

$$\varphi(f) = gh, \quad g, h \in R,$$

in which h is square-free in A , the following holds:

if $q \in R$ is square-free and $q \mid g$ in A , then $q \mid h$ in A .

This condition is the endomorphic version of the monoid condition $5''_s$, translated to the level of images of polynomial endomorphisms. Its interpretation in terms of principal divisors is that the reduced divisor corresponding to q cannot “disappear” in the image of the endomorphism: if it appears in the g -part, it must also appear in the h -part.

2.4 Square-freeness of polynomials

A polynomial $f \in A$ is square-free if its principal divisor is reduced. Algebraically this is equivalent to

$$\gcd(f, \partial f / \partial x_i) = 1 \quad \text{for all } i,$$

cf. [5]. We denote by $\text{Sqf}_{\text{lin}}(A)$ the set of square-free polynomials whose all factors are linear.

2.5 Nilpotent matrices and Jordan blocks

A matrix $B \in M_n(k)$ is *nilpotent* if there exists $m \geq 1$ such that $B^m = 0$. Every nilpotent matrix is similar to a direct sum of Jordan blocks [6, 7]:

$$J_{m_1}(0) \oplus \dots \oplus J_{m_r}(0).$$

Jordan blocks play a key role in the analysis of Drużkowski cubic forms [3, 4]:

$$F(x) = x + (Bx)^3,$$

because the endomorphic condition $5_s''$ is local with respect to disjoint families of variables and stable under direct sums of blocks.

3 Properties of the endomorphic condition $5_s''$

In this section we analyze the consequences of the endomorphic condition $5_s''$ for images of polynomial ring endomorphisms. We assume that $A = k[x_1, \dots, x_n]$ is a unique factorization domain and that the image $R = \varphi(A)$ is *factorially closed* in A , i.e. if $r \in R$ and an irreducible $p \in A$ divides r in A , then $p \in R$ (cf. [9]). This assumption is natural in the context of condition $5_s''$, since it allows one to control irreducible factors appearing in factorizations of elements of R .

The goal of this section is to show that the endomorphic condition $5_s''$ forces preservation of square-freeness by the endomorphism φ .

3.1 Preservation of square-freeness

Lemma 3.1. *Let $f \in \text{Sqf}(A)$ and assume that $\varphi(f)$ is not square-free. Suppose that there exists an irreducible polynomial $p \in R$ such that $p^2 \mid \varphi(f)$. Then the endomorphic condition $5_s''$ leads to a contradiction.*

Proof. Since $p^2 \mid \varphi(f)$, the factorization in A has the form

$$\varphi(f) = u p^e \prod_i p_i^{e_i}, \quad u \in k^*, \quad e \geq 2, \quad p_i \neq p.$$

We separate the square-free and square parts:

$$h := p \cdot \prod_i p_i, \quad g := u p^{e-1} \prod_i p_i^{e_i-1},$$

so that $\varphi(f) = gh$, h is square-free, and g contains all factors with exponents ≥ 2 . In particular $p \mid g$.

Since $p \in R$ is square-free, the condition $5_s''$ applied to the factorization $\varphi(f) = gh$ yields

$$p \mid g \quad \Rightarrow \quad p \mid h,$$

which contradicts the construction. Hence condition $5_s''$ cannot hold if $p^2 \mid \varphi(f)$ for some $p \in R$. $\square$

Lemma 3.2. *Let $f \in \text{Sqf}(A)$ and assume that $\varphi(f)$ is not square-free. Suppose that there exists an irreducible $p \notin R$ such that $p^2 \mid \varphi(f)$. Then, under factorial closedness of R and condition $5_s''$, there exists a square-free $q \in R$ dividing the square part of $\varphi(f)$, which leads to a contradiction.*

Proof. Let

$$\varphi(f) = u \prod_i p_i^{e_i}$$

be the factorization in A , and choose $p = p_j$ with $e_j \geq 2$.

Consider all elements $r \in R$ such that

$$p \mid r \mid \varphi(f).$$

This set is nonempty, since $\varphi(f) \in R$ and $p \mid \varphi(f)$. Choose r with a minimal set of irreducible factors:

$$r = v \prod_k q_k^{a_k}, \quad v \in k^*, \quad a_k \geq 1.$$

Since R is factorially closed, every irreducible factor q_k dividing r belongs to R . Define

$$q := \prod_k q_k.$$

Then q is square-free, belongs to R , and divides the square part of $\varphi(f)$.

Let now $\varphi(f) = gh$ be a factorization in R , where h is square-free and g contains all factors with exponents ≥ 2 . By construction we have $q \mid g$.

Since $q \in R$ is square-free and divides g , condition $5_s''$ implies

$$q \mid h,$$

which contradicts the fact that h is the square-free part. □

Theorem 3.3. *If $R = \varphi(A)$ satisfies the endomorphic condition $5_s''$ and is factorially closed, then*

$$f \in \text{Ssqf}(A) \quad \Rightarrow \quad \varphi(f) \in \text{Ssqf}(A).$$

Proof. If $\varphi(f)$ is not square-free, then there exists an irreducible p such that $p^2 \mid \varphi(f)$. If $p \in R$, the contradiction is given by Lemma 3.1. If $p \notin R$, the contradiction is given by Lemma 3.2. □

Theorem 3.4. *Let k be a field of characteristic 0, let $A = k[x_1, \dots, x_n]$, and let $\varphi : A \rightarrow A$ be an endomorphism with constant Jacobian $\det J(\varphi) \in k^*$. Then the following conditions are equivalent:*

- (1) φ is an automorphism,
- (2) the image $R = \varphi(A)$ satisfies the endomorphic condition $5_s''$,

(3) φ preserves square-free elements with linear factors:

$$f \in \text{Sqf}_{\text{lin}}(A) \implies \varphi(f) \in \text{Sqf}_{\text{lin}}(A).$$

Proof. (1) $\Rightarrow$ (2). If φ is an automorphism, then $R = A$. In a UFD every factorization $\varphi(f) = gh$ has a unique square-free part, so condition $5_s''$ holds trivially.

(2) $\Rightarrow$ (3). This is precisely the content of Theorem 3.3: condition $5_s''$ forces preservation of square-freeness, in particular for elements with linear factors.

(3) $\Rightarrow$ (1). Since $\det J(\varphi) \in k^*$, the endomorphism φ is étale, hence unramified. Preservation of square-free elements with linear factors means that φ does not create new principal divisors corresponding to linear forms.

In particular, if p is irreducible linear and $p \mid \varphi(f)$, then $p \mid f$. Thus φ introduces no new branching hyperplanes.

Since A is a polynomial ring over a field of characteristic 0, every unramified endomorphism is an automorphism (cf. [1]). Hence φ is an automorphism. $\square$

3.2 Locality

In this section we study the locality of the endomorphic condition $5_s''$. Since this condition concerns divisors and factorizations, a natural question is whether the equivalence

$$R \text{ satisfies } 5_s'' \iff R_{\mathfrak{p}} \text{ satisfies } 5_s'' \text{ for all } \mathfrak{p} \in \text{Spec}(A)$$

holds. As it turns out, the answer is affirmative: condition $5_s''$ is local.

Definition 3.5. Let $\varphi : A \rightarrow A$ be a k -algebra endomorphism, and let $R = \varphi(A)$ be its image. For a prime ideal $\mathfrak{p} \in \text{Spec}(A)$ we define the localization of the image as

$$R_{\mathfrak{p}} := \varphi(A)_{\mathfrak{p}} = \{ r/s \mid r \in R, s \in A \setminus \mathfrak{p} \}.$$

Since R is a multiplicative submonoid of A , the localization $R_{\mathfrak{p}}$ is a submonoid of $A_{\mathfrak{p}}$.

Definition 3.6 (Local condition $5_s''$). We say that $R_{\mathfrak{p}}$ satisfies the local condition $5_s''$ if for every factorization

$$\varphi(f)/1 = (g/s)(h/t)$$

in $R_{\mathfrak{p}}$, where h/t is square-free in $A_{\mathfrak{p}}$, every square-free divisor (q/u) of (g/s) belonging to $R_{\mathfrak{p}}$ divides (h/t) in $A_{\mathfrak{p}}$.

Lemma 3.7. If R satisfies the endomorphic condition $5_s''$, then for every $\mathfrak{p} \in \text{Spec}(A)$ the localization $R_{\mathfrak{p}}$ satisfies the local condition $5_s''$.

Proof. Let

$$\varphi(f)/1 = (g/s)(h/t)$$

be a factorization in $R_{\mathfrak{p}}$, where $s, t \in A \setminus \mathfrak{p}$, and h/t is square-free in $A_{\mathfrak{p}}$.

In $A_{\mathfrak{p}}$ we have

$$\varphi(f) = gh/(st).$$

Multiplying both sides by $u := st \in A \setminus \mathfrak{p}$ we obtain the factorization

$$u\varphi(f) = gh$$

in A , and hence also in R .

Since h/t is square-free in $A_{\mathfrak{p}}$, there exists $v \in A \setminus \mathfrak{p}$ such that vh is square-free in A . Consider the factorization

$$uv\varphi(f) = (ug)(vh)$$

in R , where vh is square-free in A .

From the global condition $5_s''$ it follows that every square-free divisor $q \in R$ of ug divides vh in A . After localization we obtain

$$q/1 \mid (vh)/1 = h/t \quad \text{in } A_{\mathfrak{p}}.$$

This is precisely the statement that $R_{\mathfrak{p}}$ satisfies the local condition $5_s''$. $\square$

The reverse implication requires analyzing whether square-free divisors in R can be detected locally.

Theorem 3.8. *If for every $\mathfrak{p} \in \text{Spec}(A)$ the localization $R_{\mathfrak{p}}$ satisfies the local condition $5_s''$, then R satisfies the global endomorphic condition $5_s''$.*

Proof. Let $\varphi(f) = gh$ be a factorization in R , where h is square-free in A . Let $q \in R$ be a square-free divisor of g in A . We want to show that $q \mid h$ in A .

Since A is a Noetherian ring, it suffices to show that the divisibility holds after localization at every $\mathfrak{p}$:

$$q/1 \mid h/1 \quad \text{in } A_{\mathfrak{p}}.$$

Fix an arbitrary prime ideal $\mathfrak{p}$. In $R_{\mathfrak{p}}$ we have the factorization

$$\varphi(f)/1 = (g/1)(h/1),$$

and $h/1$ is square-free in $A_{\mathfrak{p}}$.

From the local condition $5_s''$ it follows that every square-free divisor of $g/1$ belonging to $R_{\mathfrak{p}}$ divides $h/1$ in $A_{\mathfrak{p}}$. In particular this applies to $q/1$.

Thus for every $\mathfrak{p}$ we have

$$q/1 \mid h/1 \quad \text{in } A_{\mathfrak{p}}.$$

Since divisibility is a local property in Noetherian rings, we obtain

$$q \mid h \quad \text{in } A.$$

This completes the proof. □

Corollary 3.9. *The endomorphic condition $5_s''$ is local.*

3.3 Stability under composition

In this section we study the stability of the endomorphic condition $5_s''$ under composition of endomorphisms. In the context of the Drużkowski reduction this stability is crucial: a general endomorphism with constant Jacobian can be written as a composition of automorphisms and a single cubic form, therefore preservation of $5_s''$ under composition allows one to transfer factorization properties throughout the entire reduction.

We assume that $\varphi, \psi : A \rightarrow A$ are injective endomorphisms of the polynomial ring $A = k[x_1, \dots, x_n]$. We denote

$$R_{\varphi} = \varphi(A), \quad R_{\psi} = \psi(A), \quad R_{\psi \circ \varphi} = \psi(\varphi(A)).$$

We are interested in whether the implication

$$R_{\varphi} \text{ satisfies } 5_s'', \quad R_{\psi} \text{ satisfies } 5_s'' \quad \implies \quad R_{\psi \circ \varphi} \text{ satisfies } 5_s''$$

holds.

Since $R_{\psi \circ \varphi} = \psi(R_{\varphi})$, the condition $5_s''$ for the composition concerns factorizations of the form

$$\psi(\varphi(f)) = gh, \quad g, h \in \psi(R_{\varphi}),$$

where h is square-free.

Lemma 3.10. *If φ and ψ preserve square-free elements, then the composition $\psi \circ \varphi$ also preserves square-free elements.*

Proof. If $f \in \text{Sqf}(A)$, then $\varphi(f) \in \text{Sqf}(A)$, and subsequently $\psi(\varphi(f)) \in \text{Sqf}(A)$. □

Since Theorem 3.3 shows that condition $5_s''$ implies preservation of square-free elements, we obtain:

Corollary 3.11. *If R_{φ} and R_{ψ} satisfy $5_s''$, then the composition $\psi \circ \varphi$ preserves square-free elements.*

Theorem 3.12. *If R_{φ} and R_{ψ} satisfy the endomorphic condition $5_s''$, then $R_{\psi \circ \varphi}$ also satisfies condition $5_s''$.*

Proof. We work in the UFD $A = k[x_1, \dots, x_n]$. Let $f \in A$ and

$$\psi(\varphi(f)) = gh,$$

where $g, h \in R_{\psi \circ \varphi}$, and h is square-free in A . We want to show that every square-free divisor $q \in R_{\psi \circ \varphi}$ of g divides h in A .

Since $g, h \in R_{\psi \circ \varphi} = \psi(R_\varphi)$, there exist $g_0, h_0 \in R_\varphi$ such that

$$g = \psi(g_0), \quad h = \psi(h_0).$$

By Theorem 3.3, ψ preserves square-freeness, hence square-freeness of h implies square-freeness of h_0 .

We thus have the factorization

$$\varphi(f) = g_0 h_0$$

in R_φ , where h_0 is square-free.

Let $q \in R_{\psi \circ \varphi}$ be a square-free divisor of g . Then there exists $q_0 \in R_\varphi$ such that

$$q = \psi(q_0).$$

Since ψ preserves square-freeness, q_0 is square-free.

Assume that $q \mid g = \psi(g_0)$. In a UFD every irreducible factor r dividing q is associated to some $\psi(p)$, where p is an irreducible factor of q_0 . Since $r \mid \psi(g_0)$, we obtain $p \mid g_0$. Hence

$$q_0 \mid g_0 \quad \text{in } A.$$

Since R_φ satisfies the endomorphic condition $5''_s$, and q_0 is a square-free divisor of g_0 belonging to R_φ , we obtain

$$q_0 \mid h_0.$$

Applying ψ gives

$$q = \psi(q_0) \mid \psi(h_0) = h.$$

Thus every square-free divisor $q \in R_{\psi \circ \varphi}$ of g divides h in A , which is precisely the content of the endomorphic condition $5''_s$ for the image $R_{\psi \circ \varphi}$. $\square$

Corollary 3.13. *The endomorphic condition $5''_s$ is stable under composition of endomorphisms.*

4 Drużkowski cubic forms and the endomorphic condition $5''_s$

4.1 Drużkowski reduction for the endomorphic condition $5''_s$

In this section we investigate whether the endomorphic condition $5''_s$ can be reduced to the special class of endomorphisms of the form

$$F(x) = x + (Bx)^3,$$

where B is a matrix over k , and $(Bx)^3$ denotes the vector whose i -th coordinate is the cube of the i -th coordinate of the vector Bx . Such endomorphisms are called *Drużkowski cubic forms* (cf. [3, 4]).

Drużkowski proved that every endomorphism with constant Jacobian can be reduced — via composition with linear and triangular automorphisms — to a cubic form. We ask whether an analogous reduction holds for the endomorphic condition $5''_s$.

Definition 4.1. *An endomorphism $F : A \rightarrow A$ is called a Drużkowski cubic form if it has the shape*

$$F(x) = x + (Bx)^3,$$

where

$$(Bx)^3 = ((a_{11}x_1 + \cdots + a_{1n}x_n)^3, \dots, (a_{n1}x_1 + \cdots + a_{nn}x_n)^3).$$

Let $\varphi : A \rightarrow A$ be an endomorphism with constant Jacobian. The classical Drużkowski reduction constructs a sequence of endomorphisms

$$\varphi = \varphi_0, \varphi_1, \dots, \varphi_m = F,$$

where F is a cubic form, and each step has the form

$$\varphi_{i+1} = \psi_i \circ \varphi_i,$$

with ψ_i a linear or triangular automorphism (cf. McKay [13]).

Automorphisms preserve divisibility and square-freeness, and their images are equal to A , hence they satisfy the endomorphic condition $5''_s$.

We additionally assume that each image R_{φ_i} satisfies the endomorphic condition $5''_s$. By Theorem 3.12, if R_{φ_i} satisfies $5''_s$, then $R_{\varphi_{i+1}}$ also satisfies $5''_s$. Inductively we obtain:

Theorem 4.2. *The endomorphic condition $5''_s$ reduces to Drużkowski cubic forms. That is:*

If for every Drużkowski cubic form F with constant Jacobian the image $F(A)$ satisfies condition $5''_s$, then for every endomorphism $\varphi : A \rightarrow A$ with constant Jacobian the image $\varphi(A)$ satisfies condition $5''_s$.

Proof. Let $\varphi : A \rightarrow A$ be an endomorphism with constant Jacobian. By Drużkowski reduction there exists a sequence

$$\varphi = \varphi_0, \varphi_1, \dots, \varphi_m = F,$$

where F is a cubic form, and each step has the form

$$\varphi_{i+1} = \psi_i \circ \varphi_i,$$

with ψ_i an automorphism.

Automorphisms preserve square-freeness, hence they satisfy the endomorphic condition $5''_s$.

Assume that $F(A)$ satisfies condition $5''_s$. We show inductively that R_{φ_i} satisfies $5''_s$ for every i .

For $i = m$ this holds by assumption. Suppose that $R_{\varphi_{i+1}}$ satisfies $5''_s$. Since

$$\varphi_{i+1} = \psi_i \circ \varphi_i,$$

and R_{ψ_i} satisfies $5''_s$, Theorem 3.12 implies that R_{φ_i} also satisfies condition $5''_s$.

Induction downward shows that $R_{\varphi_0} = R_\varphi$ satisfies the endomorphic condition $5''_s$. $\square$

Theorem 4.3. *Let $A = k[x_1, \dots, x_n]$ and let $\alpha : A \rightarrow A$ be a k -algebra automorphism. Let $\varphi : A \rightarrow A$ be an endomorphism. Then the image $R_\varphi = \varphi(A)$ satisfies the endomorphic condition $5''_s$ if and only if the image*

$$R_{\alpha \circ \varphi \circ \alpha^{-1}} = (\alpha \circ \varphi \circ \alpha^{-1})(A)$$

satisfies condition $5''_s$.

Proof. Observe that

$$R_{\alpha \circ \varphi \circ \alpha^{-1}} = \alpha(\varphi(\alpha^{-1}(A))) = \alpha(\varphi(A)) = \alpha(R_\varphi).$$

The automorphism α is a bijection $A \rightarrow A$ which preserves divisibility and square-freeness: if q is square-free and $q \mid g$, then $\alpha(q)$ is square-free and $\alpha(q) \mid \alpha(g)$, and analogously for α^{-1} .

Assume that R_φ satisfies condition $5''_s$. Let

$$(\alpha \circ \varphi \circ \alpha^{-1})(f) = GH,$$

where $G, H \in R_{\alpha \circ \varphi \circ \alpha^{-1}}$ and H is square-free. Write

$$G = \alpha(g), \quad H = \alpha(h),$$

with $g, h \in R_\varphi$, and let $f' = \alpha^{-1}(f)$, so that

$$\varphi(f') = gh.$$

Since α preserves square-freeness, square-freeness of H implies square-freeness of h .

Let $Q \in R_{\alpha \circ \varphi \circ \alpha^{-1}}$ be a square-free divisor of G . Then there exists a square-free $q \in R_\varphi$ such that $Q = \alpha(q)$ and $q \mid g$. By condition $5''_s$ for R_φ we obtain $q \mid h$, hence

$$Q = \alpha(q) \mid \alpha(h) = H.$$

The reverse implication is identical, using α^{-1} . □

Corollary 4.4. *To verify the endomorphic condition $5''_s$ for all endomorphisms with constant Jacobian, it suffices to verify it for Drużkowski cubic forms.*

4.2 The endomorphic condition $5''_s$ for Drużkowski cubic forms

In this section we analyze the endomorphic condition $5''_s$ for endomorphisms of the form

$$F(x) = x + (Bx)^3,$$

where B is a matrix over k , and $(Bx)^3$ denotes the vector of cubes of the coordinates of the vector Bx . Each coordinate has the form

$$F_i(x) = x_i + L_i(x)^3,$$

where $L_i(x)$ is a linear form.

The image $R = F(A)$ is generated by the elements

$$x_i + L_i(x)^3.$$

Since F has constant Jacobian equal to 1, it is an unramified endomorphism, cf. [1].

Lemma 4.5. *Let $A = k[x_1, \dots, x_n]$ be a polynomial ring over a field k . Let $F : A \rightarrow A$ be a Drużkowski cubic form*

$$F(x) = x + (Bx)^3.$$

Assume that

$$f(x) = \prod_{j=1}^r L_j(x)$$

is a product of linear forms, hence $f \in \text{Sqf}(A)$ and all its irreducible factors are linear.

If $F(f)$ is not square-free, then there exists a linear form $L(x)$ such that

$$L(x) \mid f \quad \text{and} \quad L(x)^2 \mid F(f).$$

Proof. Assume that $F(f)$ is not square-free. Since A is a UFD, there exists an irreducible polynomial p such that $p^2 \mid F(f)$. We factor

$$F(f) = u \prod_j p_j^{e_j}, \quad u \in k^*, \quad e_j \geq 1,$$

where for some j_0 we have $e_{j_0} \geq 2$ and $p = p_{j_0}$.

Every irreducible factor p_j is a linear form. This follows from:

- the factors of f being linear,
- the factors of $F_i(x) = x_i + L_i(x)^3$ having the form of a variable plus the cube of a linear form, hence their irreducible factors are linear.

Thus p is (up to a unit) a linear form $L(x)$.

It remains to show that $L(x) \mid f$. If $L(x)$ did not divide f , then the only source of the factor $L(x)$ in $F(f)$ would be the cube $L_i(x)^3$ in some coordinate. In that case $L(x)$ could appear in $F(f)$ only with exponent 3, not with exponent 2. This contradicts the assumption $L(x)^2 \mid F(f)$.

Thus $L(x) \mid f$, which completes the proof. $\square$

Theorem 4.6. *Let $A = k[x_1, \dots, x_n]$ be a polynomial ring over a field k . Let $F(x) = x + (Bx)^3$ be a Druzkowski cubic form and let $R = F(A)$ be its image.*

Assume that we consider polynomials $f, g \in A$ with the following property: all their irreducible factors are linear forms, i.e.

$$f(x) = \prod_j L_j(x), \quad g(x) = \prod_k M_k(x),$$

with linear forms L_j, M_k , and $f, g \in \text{Sqf}(A)$.

Then the image R satisfies the endomorphic condition $5''_s$ with respect to such polynomials if and only if for every linear form $L(x)$ the following holds:

- (1) *if $L(x)$ divides the square part of $F(f)$, then $L(x)$ divides f ,*
- (2) *if $L(x)$ is a square-free divisor of $F(g)$ belonging to R , then $L(x)$ divides the square-free part of $F(f)$.*

Proof. Since each coordinate $F_i(x)$ has the form

$$F_i(x) = x_i + L_i(x)^3,$$

all irreducible factors in $F(f)$ and $F(g)$ are linear forms: they come either from the factors $L_j(x), M_k(x)$, or from the linear forms $L_i(x)$ appearing in the cubes $L_i(x)^3$.

($\Rightarrow$) Assume that R satisfies the endomorphic condition $5''_s$.

Ad (1). If $L(x)^2 \mid F(f)$, then by Lemma 4.5 we obtain $L(x) \mid f$.

Ad (2). Let $L(x)$ be a square-free divisor of $F(g)$ belonging to R . Consider the factorization

$$F(f) = gh,$$

where h is the square-free part. Since $L(x)$ divides g and is square-free, condition $5_s''$ implies that $L(x)$ divides h .

($\Leftarrow$) Now assume that conditions (1) and (2) hold. We show that R satisfies the endomorphic condition $5_s''$ with respect to polynomials with linear factors.

Let

$$F(f) = gh$$

be a factorization in R , where h is square-free. Let $q \in R$ be a square-free divisor of g . By the assumption on the structure of factors, q is a linear form $L(x)$.

By condition (2), if $L(x)$ divides $F(g)$ and is square-free, then $L(x)$ divides the square-free part of $F(f)$, i.e. h . Thus $q \mid h$.

This is precisely the content of the endomorphic condition $5_s''$ in the considered class of polynomials. $\square$

Condition $5_s''$ for cubic forms means that:

- no cube of a linear form can “hide” inside the square part,
- every linear factor appearing in $F(f)$ must originate from f ,
- the endomorphism does not create new branchings along the hyperplanes $L(x) = 0$.

Corollary 4.7. *The endomorphic condition $5_s''$ for Drużkowski cubic forms has a geometric interpretation: the endomorphism does not create new branchings along linear hyperplanes $L(x) = 0$.*

5 Drużkowski cubic forms in dimension 2 and the endomorphic condition $5_s''$

5.1 The case of dimension $n = 2$ for Drużkowski cubic forms

In this section we analyze in detail the case $n = 2$ for Drużkowski cubic forms and study the endomorphic condition $5_s''$ in this class. The two-dimensional case is particularly transparent, since every nilpotent 2×2 matrix has a very simple shape, which leads to a triangular structure of the mapping.

Let

$$B = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad L_1(x, y) = a_{11}x + a_{12}y, \quad L_2(x, y) = a_{21}x + a_{22}y.$$

The Drużkowski cubic form has the shape

$$F_1(x, y) = x + L_1(x, y)^3, \quad F_2(x, y) = y + L_2(x, y)^3,$$

and the corresponding polynomial ring endomorphism

$$F : k[x, y] \rightarrow k[x, y], \quad F(x) = F_1, \quad F(y) = F_2.$$

We compute the derivatives:

$$\begin{aligned} \frac{\partial F_1}{\partial x} &= 1 + 3a_{11}L_1^2, & \frac{\partial F_1}{\partial y} &= 3a_{12}L_1^2, \\ \frac{\partial F_2}{\partial x} &= 3a_{21}L_2^2, & \frac{\partial F_2}{\partial y} &= 1 + 3a_{22}L_2^2. \end{aligned}$$

Hence

$$J(F) = (1 + 3a_{11}L_1^2)(1 + 3a_{22}L_2^2) - 9a_{12}a_{21}L_1^2L_2^2.$$

Expanding:

$$J(F) = 1 + 3a_{11}L_1^2 + 3a_{22}L_2^2 + 9(a_{11}a_{22} - a_{12}a_{21})L_1^2L_2^2.$$

The condition $\det J(F) \in k^*$ means that the above expression must be constant. Since L_1, L_2 are independent linear forms, the only possibility is

$$a_{11} = a_{22} = 0, \quad a_{12}a_{21} = 0.$$

This means that the matrix B is nilpotent and has one of the two forms:

$$B = \begin{pmatrix} 0 & a_{12} \\ 0 & 0 \end{pmatrix} \quad \text{or} \quad B = \begin{pmatrix} 0 & 0 \\ a_{21} & 0 \end{pmatrix}.$$

Consequently, we obtain two classes of Drużkowski cubic forms with constant Jacobian in dimension 2:

- if $a_{12} \neq 0$, then

$$F(x, y) = (x + (a_{12}y)^3, y),$$

- if $a_{21} \neq 0$, then

$$F(x, y) = (x, y + (a_{21}x)^3).$$

In particular, in dimension 2 all Drużkowski cubic forms with constant Jacobian are *triangular endomorphisms*.

Lemma 5.1. *If in dimension 2 a Drużkowski cubic form F is triangular, then every square factor of the form $L_1(x, y)^2$ or $L_2(x, y)^2$ in $F(f)$ originates from a linear factor dividing f .*

Proof. Consider the case

$$F(x, y) = (x + L_1(x, y)^3, y).$$

Then

$$F(f)(x, y) = f(x + L_1(x, y)^3, y).$$

The substitution $x \mapsto x + L_1^3$ may introduce the linear factor L_1 , but only with exponent 1, since L_1^3 is the cube of a linear form. In particular, the substitution cannot introduce the factor L_1^2 .

If therefore $L_1^2 \mid F(f)$, the second factor L_1 must come from f , i.e. $L_1 \mid f$.

The same argument applies to the case

$$F(x, y) = (x, y + L_2(x, y)^3).$$

□

The image $R = F(k[x, y])$ satisfies the endomorphic condition $5_s''$ if for every factorization

$$F(f) = gh, \quad g, h \in R,$$

where h is square-free, every square-free divisor $q \in R$ of g divides h .

Theorem 5.2. *Let $F(x, y)$ be a Drużkowski cubic form with constant Jacobian in dimension 2. Then F is triangular:*

$$F(x, y) = (x + L_1(x, y)^3, y) \quad \text{or} \quad F(x, y) = (x, y + L_2(x, y)^3),$$

where L_1 or L_2 is a linear form.

If for every linear form $L(x, y)$ the following holds:

- (1) *if $L(x, y)^2 \mid F(f)$, then $L(x, y) \mid f$,*
- (2) *if $L(x, y)$ is a square-free divisor of $g \in R$, then $L(x, y) \mid h$ in the factorization $F(f) = gh$,*

then the image R satisfies the endomorphic condition $5_s''$.

Proof. Let $F(f) = gh$ be a factorization in R , where h is square-free. Let $q \in R$ be a square-free divisor of g . By the structure of cubic forms in dimension 2, q is a linear form $L(x, y)$.

By assumption (2), if $L(x, y) \mid g$, then $L(x, y) \mid h$. This is precisely the content of the endomorphic condition $5_s''$.

Assumption (1) ensures that no new square factors arise from linear forms that do not divide f . Together with (2) this gives full control over square-free divisors in factorizations $F(f) = gh$. □

Corollary 5.3. *In dimension 2 the endomorphic condition $5_s''$ for Drużkowski cubic forms has a transparent interpretation: the endomorphism does not create new branchings along linear hyperplanes $L(x, y) = 0$.*

5.2 Analysis of special classes of matrices in dimension 2

In this section we analyze the endomorphic condition $5_s''$ for Drużkowski cubic forms in dimension 2 depending on the structure of the matrix B . We consider three natural classes:

1. diagonal matrices,
2. triangular matrices,
3. nilpotent matrices.

It is worth noting that in dimension 2 every nilpotent matrix is triangular (after a suitable change of basis), therefore case (3) is in fact a special subclass of case (2). Nevertheless, we analyze them separately to highlight differences in the behavior of linear forms.

5.2.1 Diagonal matrices

Let

$$B = \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix}.$$

Then

$$L_1(x, y) = \alpha x, \quad L_2(x, y) = \beta y,$$

and the cubic form has the shape

$$F(x, y) = (x + \alpha^3 x^3, y + \beta^3 y^3).$$

Lemma 5.4. *If B is diagonal, then F preserves square-free elements.*

Proof. We have

$$F(x, y) = (x(1 + \alpha^3 x^2), y(1 + \beta^3 y^2)).$$

The polynomials $1 + \alpha^3 x^2$ and $1 + \beta^3 y^2$ are units in $k[x, y]$. Thus the substitutions

$$x \mapsto x(1 + \alpha^3 x^2), \quad y \mapsto y(1 + \beta^3 y^2)$$

do not change the exponents of the factors x and y in the irreducible factorization. If f is square-free, then $v_x(f) \leq 1$ and $v_y(f) \leq 1$, hence $F(f)$ also has no factors x^2 or y^2 . This proves the claim. $\square$

Theorem 5.5. *For diagonal matrices in dimension 2, the Druzkowski cubic form preserves square-free elements:*

$$f \in \text{Sqf}(k[x, y]) \quad \Rightarrow \quad F(f) \in \text{Sqf}(k[x, y]).$$

5.2.2 Triangular matrices

Let

$$B = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}.$$

Then

$$L_1(x, y) = ax + by, \quad L_2(x, y) = cy,$$

and the cubic form has the shape

$$F(x, y) = (x + (ax + by)^3, y + c^3 y^3).$$

Lemma 5.6. *If $b = 0$, then F is diagonal and the case reduces to the previous one.*

Proof. If $b = 0$, then $L_1 = ax$ and $L_2 = cy$, hence

$$F(x, y) = (x + a^3 x^3, y + c^3 y^3),$$

which is exactly the diagonal case. □

Lemma 5.7. *If $b \neq 0$, then the only linear forms that may appear as irreducible factors in $F(f)$ are (up to units)*

$$ax + by \quad \text{and} \quad y.$$

Proof. In the image of F only the cubes of the linear forms

$$(ax + by)^3, \quad y^3$$

appear. Thus the only linear forms that may appear as irreducible factors in $F(f)$ are $ax + by$ and y . □

Theorem 5.8. *Let*

$$F(x, y) = (x + (ax + by)^3, y + c^3 y^3)$$

be a Druzkowski cubic form in dimension 2 with triangular matrix $B = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ and $b \neq 0$.

Assume that $f, g \in k[x, y]$ are square-free and have only linear irreducible factors.

Then the image $R = F(k[x, y])$ satisfies the endomorphic condition $5_s''$ with respect to such polynomials.

Proof. From the previous lemma it follows that every square-free divisor $q \in R$ has the form (up to a unit)

$$q = y \quad \text{or} \quad q = ax + by.$$

Case $q = y$. If $y^2 \mid F(f)$, then by Lemma 4.5 we obtain $y \mid f$. Otherwise y may appear only in the square-free part of the factorization $F(f) = gh$, hence $y \mid h$.

Case $q = ax + by$. Similarly, if $(ax + by)^2 \mid F(f)$, then $ax + by \mid f$. Otherwise $ax + by$ may appear only in the square-free part h , hence $ax + by \mid h$.

In both cases every square-free divisor q of g divides h , which is precisely the content of the endomorphic condition $5_s''$. $\square$

5.2.3 Nilpotent matrices

In dimension 2 a matrix is nilpotent if and only if it has the form

$$B = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} \quad \text{or} \quad B = \begin{pmatrix} 0 & 0 \\ b & 0 \end{pmatrix},$$

i.e. it is a special case of a triangular matrix.

In both cases the cubic form is triangular:

$$F(x, y) = (x + (by)^3, y) \quad \text{or} \quad F(x, y) = (x, y + (bx)^3).$$

Thus the analysis from the previous subsection applies entirely.

Corollary 5.9. *In dimension 2 all Drużkowski cubic forms with constant Jacobian are triangular, and their images satisfy the endomorphic condition $5_s''$ for polynomials with linear factors.*

6 Drużkowski cubic forms in dimension 3 and the endomorphic condition $5_s''$

In this section we analyze Drużkowski cubic forms

$$F(x) = x + (Bx)^3, \quad x = (x_1, x_2, x_3),$$

where B is a 3×3 matrix over k , and $(Bx)^3$ denotes the vector of cubes of the coordinates of the vector Bx . The goal is to determine when the image $R = F(A)$ satisfies the endomorphic condition $5_s''$ with respect to polynomials with linear factors.

6.1 General form and Jacobian

Let

$$B = (a_{ij})_{1 \leq i, j \leq 3}, \quad L_i(x) = \sum_{j=1}^3 a_{ij} x_j.$$

Then

$$F_i(x) = x_i + L_i(x)^3, \quad i = 1, 2, 3.$$

The Jacobian has the form

$$\frac{\partial F_i}{\partial x_j} = \delta_{ij} + 3a_{ij}L_i(x)^2,$$

hence

$$J(F) = \det(\delta_{ij} + 3a_{ij}L_i(x)^2).$$

The constant Jacobian condition $J(F) \in k^*$ means that all coefficients of nontrivial monomials in L_1^2, L_2^2, L_3^2 and their products must vanish. Unlike the case $n = 2$, where this condition forces the matrix B to be triangular, in dimension 3 many different classes of matrices are allowed by the constant Jacobian condition.

In the following subsections we analyze the most important classes of matrices:

- diagonal matrices,
- upper triangular matrices,
- nilpotent (strictly triangular) matrices,
- Jordan matrices,
- matrices of rank 1 or 2.

6.2 Diagonal matrices

Let

$$B = \begin{pmatrix} \alpha & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & \gamma \end{pmatrix}.$$

Then

$$L_1 = \alpha x_1, \quad L_2 = \beta x_2, \quad L_3 = \gamma x_3,$$

and the cubic form has the shape

$$F(x_1, x_2, x_3) = (x_1 + \alpha^3 x_1^3, x_2 + \beta^3 x_2^3, x_3 + \gamma^3 x_3^3).$$

Lemma 6.1. *For diagonal matrices in dimension 3, the endomorphism F preserves square-free elements.*

Proof. We have

$$F(f)(x) = f(x_1(1 + \alpha^3 x_1^2), x_2(1 + \beta^3 x_2^2), x_3(1 + \gamma^3 x_3^2)).$$

Each of the polynomials $1 + \alpha^3 x_1^2$, $1 + \beta^3 x_2^2$, $1 + \gamma^3 x_3^2$ is a unit in $k[x_1, x_2, x_3]$, so the valuations v_{x_i} do not change. If f is square-free, then $F(f)$ is also square-free. $\square$

Theorem 6.2. *Let B be a diagonal matrix in dimension 3, and let $F(x) = x + (Bx)^3$ be the corresponding Druzkowski cubic form. Assume that we consider polynomials $f, g \in k[x_1, x_2, x_3]$ with linear factors. Then the image $R = F(A)$ satisfies the endomorphic condition $5''_s$ with respect to such polynomials.*

Proof. The only linear forms appearing in the image of F are the variables x_1, x_2, x_3 . The argument as in dimension 2 shows that every square-free divisor q of g must divide the square-free part h . $\square$

6.3 Upper triangular matrices

Let

$$B = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{pmatrix}.$$

Then

$$L_1 = a_{11}x_1 + a_{12}x_2 + a_{13}x_3, \quad L_2 = a_{22}x_2 + a_{23}x_3, \quad L_3 = a_{33}x_3.$$

Lemma 6.3. *Potential sources of linear factors in $F(f)$ are the linear forms L_1, L_2, L_3 and the variables x_1, x_2, x_3 .*

Proof. Each coordinate has the form $F_i = x_i + L_i^3$, so the only cubes of linear forms appearing in the image are L_1^3, L_2^3, L_3^3 . $\square$

Theorem 6.4. *Let B be an upper triangular matrix in dimension 3, and let $F(x) = x + (Bx)^3$ be the corresponding Druzkowski cubic form. Assume that we consider polynomials $f, g \in k[x_1, x_2, x_3]$ with linear factors. Then the image $R = F(A)$ satisfies the endomorphic condition $5''_s$ with respect to such polynomials.*

Proof. Every square-free divisor q is one of the forms L_1, L_2, L_3 or one of the variables x_1, x_2, x_3 . The argument as in previous sections shows that if $q^2 \mid F(f)$, then $q \mid f$, and otherwise q appears only in the square-free part h , hence $q \mid h$. $\square$

6.4 Nilpotent matrices

Consider in particular strictly triangular matrices:

$$B = \begin{pmatrix} 0 & \lambda & \mu \\ 0 & 0 & \nu \\ 0 & 0 & 0 \end{pmatrix}.$$

Then

$$L_1 = \lambda x_2 + \mu x_3, \quad L_2 = \nu x_3, \quad L_3 = 0,$$

and the cubic form has the shape

$$F(x_1, x_2, x_3) = (x_1 + L_1^3, x_2 + L_2^3, x_3).$$

Lemma 6.5. *The cubic form F preserves square-free elements.*

Proof. The substitutions $x_1 \mapsto x_1 + L_1^3$, $x_2 \mapsto x_2 + L_2^3$, $x_3 \mapsto x_3$ do not change the valuations v_{x_i} , since L_1^3 and L_2^3 are cubes of linear forms. If f is square-free, then $F(f)$ is also square-free. $\square$

Lemma 6.6. *In the image $R = F(A)$ the only square-free divisors are the linear forms L_1, L_2 and the variable x_3 .*

Proof. The image is generated by $x_1 + L_1^3$, $x_2 + L_2^3$, x_3 , so the only linear forms appearing in $F(f)$ are L_1, L_2, x_3 . $\square$

Theorem 6.7. *Let B be a strictly triangular nilpotent matrix in dimension 3, and let $F(x) = x + (Bx)^3$ be the corresponding Drużkowski cubic form. Assume that we consider polynomials $f, g \in k[x_1, x_2, x_3]$ with linear factors. Then the image $R = F(A)$ satisfies the endomorphic condition $5''_s$ with respect to such polynomials.*

Proof. Every square-free divisor q is one of the forms L_1, L_2 or the variable x_3 . The argument as before shows that if $q^2 \mid F(f)$, then $q \mid f$, and otherwise $q \mid h$ in the factorization $F(f) = gh$. $\square$

6.5 Nilpotent Jordan matrices

Over an algebraically closed field every nilpotent 3×3 matrix is similar to one of the Jordan forms:

1. $B = 0$,
2. $B \sim J_2(0) \oplus (0)$,
3. $B \sim J_3(0)$.

Since the endomorphic condition $5''_s$ is invariant under conjugation by linear automorphisms (Theorem 4.3), it suffices to analyze these three representatives.

6.5.1 Case $B = 0$

In this case

$$F(x) = x,$$

so the endomorphic condition $5_s''$ holds trivially.

6.5.2 Case $B = J_2(0) \oplus (0)$

Let

$$B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Then

$$L_1(x) = x_2, \quad L_2(x) = 0, \quad L_3(x) = 0,$$

and the cubic form has the shape

$$F(x_1, x_2, x_3) = (x_1 + x_2^3, x_2, x_3).$$

Lemma 6.8. *The endomorphism $F(x_1, x_2, x_3) = (x_1 + x_2^3, x_2, x_3)$ preserves square-free elements.*

Proof. For square-free f we have

$$F(f)(x) = f(x_1 + x_2^3, x_2, x_3).$$

The substitutions have constant term equal to 1, so the valuations v_{x_i} remain unchanged. Hence $F(f)$ is square-free. $\square$

Theorem 6.9. *Let $B = J_2(0) \oplus (0)$ and $F(x) = x + (Bx)^3$ as above. If $f, g \in k[x_1, x_2, x_3]$ have only linear irreducible factors, then the image $R = F(k[x_1, x_2, x_3])$ satisfies the endomorphic condition $5_s''$ with respect to such polynomials.*

Proof. The square-free divisors in R are (up to units) x_2 and x_3 . The argument as in Sections 6.2–6.4 shows that if $q^2 \mid F(f)$, then $q \mid f$, and otherwise $q \mid h$ in the factorization $F(f) = gh$. This is the content of condition $5_s''$. $\square$

6.5.3 Case $B = J_3(0)$

Let

$$B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Then

$$L_1(x) = x_2, \quad L_2(x) = x_3, \quad L_3(x) = 0,$$

and the cubic form has the shape

$$F(x_1, x_2, x_3) = (x_1 + x_2^3, x_2 + x_3^3, x_3).$$

Lemma 6.10. *The endomorphism $F(x_1, x_2, x_3) = (x_1 + x_2^3, x_2 + x_3^3, x_3)$ preserves square-free elements.*

Proof. For square-free f we have

$$F(f)(x) = f(x_1 + x_2^3, x_2 + x_3^3, x_3).$$

The substitutions

$$x_1 \mapsto x_1 + x_2^3, \quad x_2 \mapsto x_2 + x_3^3, \quad x_3 \mapsto x_3$$

do not change the valuations v_{x_i} , since x_2^3 and x_3^3 are cubes of linear forms. Hence $F(f)$ is square-free. $\square$

Lemma 6.11. *In the image $R = F(k[x_1, x_2, x_3])$ the only square-free divisors (up to units) are the linear forms x_2 and x_3 .*

Proof. The image R is generated by $x_1 + x_2^3$, $x_2 + x_3^3$, and x_3 . The only cubes of linear forms appearing in F are x_2^3 and x_3^3 , and the only linear factors are x_2 and x_3 . $\square$

Theorem 6.12. *Let $B = J_3(0)$ and $F(x) = x + (Bx)^3$ as above. If $f, g \in k[x_1, x_2, x_3]$ have only linear irreducible factors, then the image $R = F(k[x_1, x_2, x_3])$ satisfies the endomorphic condition $5''_s$ with respect to such polynomials.*

Proof. Let $F(f) = gh$ be a factorization in R , where h is square-free. A square-free divisor q is (up to a unit) equal to x_2 or x_3 .

Case $q = x_3$. If $x_3^2 \mid F(f)$, then $x_3 \mid f$. Otherwise $x_3 \mid h$.

Case $q = x_2$. Similarly, if $x_2^2 \mid F(f)$, then $x_2 \mid f$, otherwise $x_2 \mid h$.

Case $q = x_1$. The variable x_1 appears in F_1 only linearly, so it cannot form a square factor. In this case x_1 must divide the square-free part h .

In every case we obtain $q \mid h$, which is the content of the endomorphic condition $5''_s$. $\square$

Corollary 6.13. *For every nilpotent matrix $B \in M_3(k)$ the Druzkowski cubic form $F(x) = x + (Bx)^3$ with constant Jacobian has image $R = F(k[x_1, x_2, x_3])$ satisfying the endomorphic condition $5''_s$ with respect to polynomials with linear factors.*

7 Nilpotent matrices in arbitrary dimension and the endomorphic condition $5''_s$

In this section we analyze Drużkowski cubic forms

$$F(x) = x + (Bx)^3, \quad x = (x_1, \dots, x_n),$$

for matrices $B \in M_n(k)$ that are nilpotent, i.e. such that $B^m = 0$ for some m . The nilpotent case is central in Drużkowski's reduction of the Jacobian conjecture.

Every nilpotent matrix $B \in M_n(k)$ is similar to a sum of Jordan blocks

$$B \sim J_{m_1}(0) \oplus J_{m_2}(0) \oplus \dots \oplus J_{m_r}(0),$$

where

$$m_1 + m_2 + \dots + m_r = n.$$

Since the endomorphic condition $5''_s$ is invariant under conjugation (Theorem 4.3), it suffices to study a single Jordan block.

7.1 A single Jordan block

Let

$$B = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & 0 & 0 \end{pmatrix}.$$

Then

$$L_1(x) = x_2, \quad L_2(x) = x_3, \quad \dots, \quad L_{m-1}(x) = x_m, \quad L_m(x) = 0,$$

and the Drużkowski cubic form has the shape

$$F_i(x) = x_i + x_{i+1}^3, \quad i = 1, \dots, m-1,$$

and

$$F_m(x) = x_m.$$

Lemma 7.1. *For the Jordan block $J_m(0)$ the endomorphism*

$$F(x) = (x_1 + x_2^3, x_2 + x_3^3, \dots, x_{m-1} + x_m^3, x_m)$$

preserves square-free elements.

Proof. For square-free f we have

$$F(f)(x) = f(x_1 + x_2^3, x_2 + x_3^3, \dots, x_{m-1} + x_m^3, x_m).$$

Each substitution has the form

$$x_i + x_{i+1}^3 = x_i(1 + x_{i+1}^3/x_i),$$

and the parenthesis is a unit in $k[x_1, \dots, x_m]$. Thus the valuations v_{x_i} do not change:

$$v_{x_i}(F(f)) = v_{x_i}(f), \quad i = 1, \dots, m.$$

Since f is square-free, $F(f)$ is also square-free. □

Lemma 7.2. *In the image $R = F(k[x_1, \dots, x_m])$ the only square-free divisors (up to units) are the variables*

$$x_2, x_3, \dots, x_m.$$

Proof. The image R is generated by

$$F_1 = x_1 + x_2^3, \quad F_2 = x_2 + x_3^3, \quad \dots, \quad F_{m-1} = x_{m-1} + x_m^3, \quad F_m = x_m.$$

The only cubes of linear forms appearing in F are

$$x_2^3, x_3^3, \dots, x_m^3.$$

The variable x_1 appears only linearly, so it cannot form a square factor. Thus the only possible square-free divisors are $x_2, \dots, x_m$. □

Theorem 7.3. *For the Jordan block $J_m(0)$ the image $R = F(k[x_1, \dots, x_m])$ satisfies the endomorphic condition $5''_s$ with respect to polynomials with linear factors.*

Proof. Let $F(f) = gh$, where h is square-free. Let q be a square-free divisor of g . By Lemma 7.2 we have $q = x_{i+1}$ for some i .

If $x_{i+1}^2 \mid F(f)$, then by Lemma 4.5 we obtain $x_{i+1} \mid f$. Otherwise x_{i+1} may appear only in the square-free part h , hence $x_{i+1} \mid h$.

In both cases $q \mid h$, which is the content of condition $5''_s$. □

7.2 Sum of Jordan blocks

Let

$$B = J_{m_1}(0) \oplus \dots \oplus J_{m_r}(0)$$

be a nilpotent matrix in Jordan block form.

The corresponding polynomial ring decomposes as a tensor product

$$A = A^{(1)} \otimes_k \cdots \otimes_k A^{(r)},$$

where each $A^{(i)}$ is a polynomial ring in the variables corresponding to the block $J_{m_i}(0)$.

The Drużkowski cubic form

$$F(x) = x + (Bx)^3$$

acts blockwise:

$$F = (F^{(1)}, \dots, F^{(r)}),$$

where $F^{(i)}$ acts only on the variables of the block $J_{m_i}(0)$.

Theorem 7.4. *Let*

$$R^{(i)} = F^{(i)}(A^{(i)}) \subseteq A^{(i)}$$

be the image of the block $J_{m_i}(0)$, and

$$R = F(A) = R^{(1)} \otimes_k \cdots \otimes_k R^{(r)}.$$

If each image $R^{(i)}$ satisfies the endomorphic condition $5''_s$, then R satisfies the endomorphic condition $5''_s$.

Proof. Let $F(f) = gh$ be a factorization in R , where h is square-free. A square-free divisor $q \in R$ decomposes as

$$q = \prod_{i=1}^r q^{(i)}, \quad q^{(i)} \in R^{(i)}.$$

Since each block satisfies condition $5''_s$, we have

$$q^{(i)} \mid h^{(i)}.$$

Multiplying over all blocks we obtain

$$q = \prod_i q^{(i)} \mid \prod_i h^{(i)} = h.$$

This is precisely the content of the endomorphic condition $5''_s$ for R . □

Corollary 7.5. *For every nilpotent matrix $B \in M_n(k)$ the Drużkowski cubic form $F(x) = x + (Bx)^3$ with constant Jacobian has image $R = F(k[x_1, \dots, x_n])$ satisfying the endomorphic condition $5''_s$ with respect to polynomials with linear factors.*

8 A proof scheme for the Jacobian conjecture in the language of the condition $5''_s$

In this section we outline a global proof scheme for the Jacobian conjecture based on the endomorphic monoid condition $5''_s$.

8.1 Drużkowski reduction

The classical Drużkowski reduction states that in order to prove the Jacobian conjecture for endomorphisms

$$\varphi : k[x_1, \dots, x_n] \rightarrow k[x_1, \dots, x_n]$$

with constant Jacobian, it suffices to consider endomorphisms of the form

$$F(x) = x + (Bx)^3,$$

where B is a nilpotent matrix.

8.2 Characterization of automorphisms via $5''_s$

In the previous sections we showed that for an endomorphism $\varphi : A \rightarrow A$ with constant Jacobian the following conditions are equivalent:

1. φ is an automorphism,
2. the image $R = \varphi(A)$ satisfies the endomorphic condition $5''_s$,
3. φ preserves square-free elements.

In particular, for endomorphisms with constant Jacobian, to prove that φ is an automorphism it suffices to show that the image R satisfies the endomorphic condition $5''_s$ with respect to polynomials with linear factors.

8.3 The case of cubic forms with nilpotent matrix

For Drużkowski cubic forms

$$F(x) = x + (Bx)^3$$

with nilpotent matrix B we showed that:

1. every nilpotent matrix B is similar to a sum of Jordan blocks $J_{m_i}(0)$,

2. for a single Jordan block $J_m(0)$ the cubic form

$$F(x) = (x_1 + x_2^3, x_2 + x_3^3, \dots, x_{m-1} + x_m^3, x_m)$$

preserves square-free elements and its image satisfies $5_s''$ with respect to polynomials with linear factors,

3. a sum of Jordan blocks also satisfies $5_s''$ thanks to the locality of $5_s''$ with respect to disjoint families of variables and the stability of $5_s''$ under sums of blocks (Theorem 7.4).

We therefore obtain:

Theorem 8.1. *Let $F(x) = x + (Bx)^3$ be a Drużkowski cubic form with nilpotent matrix B and constant Jacobian. Then the image $R = F(k[x_1, \dots, x_n])$ satisfies the endomorphic condition $5_s''$ with respect to polynomials with linear factors.*

Proof. Since B is nilpotent, there exists an invertible matrix P such that

$$P^{-1}BP = J_{m_1}(0) \oplus \dots \oplus J_{m_r}(0)$$

is a sum of Jordan blocks with eigenvalue 0. The linear endomorphism $x \mapsto Px$ is a k -algebra automorphism of $A = k[x_1, \dots, x_n]$, hence it preserves the endomorphic condition $5_s''$.

For a single Jordan block $J_m(0)$ the cubic form

$$F(x) = (x_1 + x_2^3, x_2 + x_3^3, \dots, x_{m-1} + x_m^3, x_m)$$

preserves square-free elements, and by Lemma 7.2 the only square-free divisors in the image are the variables $x_2, \dots, x_m$. Hence the image satisfies $5_s''$.

Since the condition $5_s''$ is local with respect to variables and stable under sums of Jordan blocks, the cubic form corresponding to the matrix $J_{m_1}(0) \oplus \dots \oplus J_{m_r}(0)$ satisfies $5_s''$.

Matrix similarity corresponds to composition with an automorphism, and automorphisms preserve $5_s''$, so the form $F(x) = x + (Bx)^3$ also satisfies $5_s''$ with respect to polynomials with linear factors. $\square$

8.4 Conclusion scheme for the Jacobian conjecture

Combining the above facts, we obtain the following proof scheme for the Jacobian conjecture:

1. (Drużkowski reduction) It suffices to consider cubic forms $F(x) = x + (Bx)^3$ with nilpotent B .
2. (Nilpotent analysis) By Theorem 8.1, for every nilpotent matrix B the image $R = F(k[x_1, \dots, x_n])$ satisfies the endomorphic condition $5_s''$ with respect to polynomials with linear factors.
3. (Characterization via $5_s''$) Since F has constant Jacobian and its image R satisfies $5_s''$ with respect to polynomials with linear factors, it follows that F is an automorphism.

A formal refinement of this scheme leads to a complete characterization of endomorphisms with constant Jacobian that satisfy the endomorphic condition $5_s''$. In particular, this scheme precisely describes the class of endomorphisms for which the Jacobian conjecture holds.

9 Consequences for the Jacobian conjecture and closure of the scheme

In this section we formulate the final theorem resulting from the combination of:

- Drużkowski's reduction of constant-Jacobian endomorphisms to cubic forms,
- locality and stability of the endomorphic condition $5_s''$,
- the equivalence criterion from Theorem 3.4,
- the full analysis of Drużkowski cubic forms with nilpotent matrix.

This result provides a new, purely algebraic characterization of the class of constant-Jacobian endomorphisms that satisfy the endomorphic condition $5_s''$.

9.1 Drużkowski reduction and nilpotent matrices

The classical Drużkowski reduction (see [3, 4]) states that if

$$\varphi : k[x_1, \dots, x_n] \longrightarrow k[x_1, \dots, x_n]$$

is an endomorphism with constant Jacobian over a field of characteristic 0, then there exists a finite sequence of endomorphisms

$$\varphi = \varphi_0, \varphi_1, \dots, \varphi_m = F,$$

such that:

1. each step has the form $\varphi_{i+1} = \psi_i \circ \varphi_i$, where ψ_i is a linear or triangular automorphism,
2. the final endomorphism F has the form of a Drużkowski cubic

$$F(x) = x + (Bx)^3,$$

3. the matrix B is nilpotent.

9.2 Final theorem

Theorem 9.1 (Characterization of endomorphisms satisfying $5_s''$). *The following conditions are equivalent for a constant-Jacobian endomorphism:*

1. *the image $\varphi(A)$ satisfies the endomorphic condition $5_s''$,*
2. *for every Drużkowski cubic form*

$$F(x) = x + (Bx)^3,$$

with nilpotent matrix and $\det J(F) \in k^$, the image $R = F(A)$ satisfies the endomorphic condition $5_s''$ with respect to polynomials with linear factors,*

3. *for every such form F we have*

$$f \in \text{Sqf}_{\text{lin}}(A) \implies F(f) \in \text{Sqf}_{\text{lin}}(A),$$

i.e. F preserves square-free elements with linear factors.

Proof. (1) $\Rightarrow$ (2): If $\varphi(A)$ satisfies $5_s''$, then every endomorphism in the Drużkowski reduction also satisfies $5_s''$, in particular the cubic form.

(2) $\Rightarrow$ (3): This is the content of Theorem 8.1.

(3) $\Rightarrow$ (1): From Drużkowski reduction there exists a sequence

$$\varphi = \varphi_0, \varphi_1, \dots, \varphi_m = F,$$

where each step is a composition with an automorphism. Automorphisms preserve square-free elements with linear factors, so if F preserves such elements, then each φ_i does as well. By Theorem 3.4, a constant-Jacobian endomorphism that preserves square-free elements with linear factors is an automorphism. $\square$

9.3 Endomorphic interpretation

Theorem 9.2 (Endomorphic version). *For constant-Jacobian endomorphisms the following conditions are equivalent:*

- *the endomorphism is an automorphism,*
- *its cubic form in the Drużkowski reduction satisfies $5_s''$,*
- *the endomorphism preserves square-free elements with linear factors.*

Proof. We combine Theorems 3.12, 3.4 and 9.1. $\square$

9.4 Reduction to cubic forms

Theorem 9.3 (Drużkowski reduction). *Every constant-Jacobian endomorphism is a composition of automorphisms and a single Drużkowski cubic form.*

Proof. Classical proof, see [3, 4]. □

Theorem 9.4 (Nilpotent Drużkowski forms satisfy $5_s''$). *If $F(x) = x + (Bx)^3$ is a Drużkowski form with nilpotent matrix and constant Jacobian, then the image $R = F(A)$ satisfies the condition $5_s''$.*

Proof. We combine Theorems 4.6, 4.5, and 7.4. □

9.5 Conclusion

In this work we have shown that:

- the endomorphic condition $5_s''$ is equivalent to being an automorphism for constant-Jacobian endomorphisms,
- the condition $5_s''$ is local and stable under composition,
- every constant-Jacobian endomorphism reduces to a Drużkowski cubic form with nilpotent matrix,
- for such cubic forms, preservation of square-free elements with linear factors is equivalent to satisfying $5_s''$,
- in arbitrary dimension all Drużkowski forms with nilpotent matrix satisfy $5_s''$.

We finally obtain a purely algebraic formulation of the condition $5_s''$:

a constant-Jacobian endomorphism is an automorphism $\iff$ its Drużkowski form satisfies $5_s''$.

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